problem1

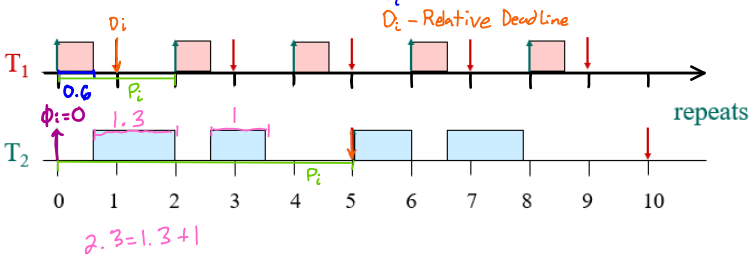
Sunday, February 1, 2026 8:45 PM

Proof of Non-tightness

To see that $\Delta > 1$ doesn't imply non-schedulability, consider the following example.

Example: We have two tasks $T_1 = (2, 0.6, 1)$ and $T_2 = (5, 2.3)$. $\Delta = 0.6/1 + 2.3/5 = 1.06$. Nonetheless, we can schedule this task set under EDF:

P_i – Period
e_i – Execution Cost
D_i – Relative Deadline



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Notation of Real-Time Tasks

- Model assumed in this chapter:
  - n periodic tasks $T_1,\ldots,T_n$.
  - The "rest of the world" periodic model is assumed.
  - T_i is specified by (ϕ_i, p_i, e_i, D_i), where
    - ϕ_i is its phase (0 unless indicated)
    - p_i is its period (an integer or, at least, rational)
    - e_i is its execution cost per job, and
    - D_i is its relative deadline.
    - Will abbreviate as (p_i,e_i,D_i) if $\phi_i=0$, and (p_i,e_i) if $\phi_i=0 \wedge p_i=D_i$.

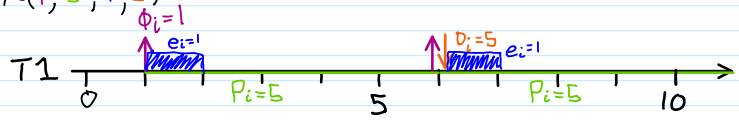
And
Not XOR

3

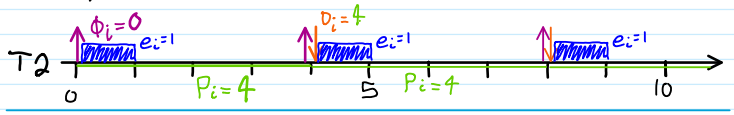
1. (10 points) Determine if the following independent task set is schedulable on a uniprocessor under EDF. Present your calculations as necessary (for the entire system / for each task) as needed. Also construct a Gantt chart to show the timeline of execution per job from time 0 to 25 and indicate preemptions, if any. T1 = (1, 5, 1, 5), T2 = (4, 1), T3 = (10, 4, 8)

ϕ_i – Phase – Time after 0 Arrow before task starts to execution (Orange Arrow)
P_i – Period – Duration between two Purple Arrows
e_i – Execution Cost – Task Duration
D_i – Relative Deadline – Time after Purple Arrow where Orange Arrow should be Placed

$T=(\phi_i, P_i, e_i, D_i)$
$T=(P_i, e_i, D_i) \rightarrow \phi_i=0$
$T=(P_i, e_i) \rightarrow \phi_i=0$ & $p_i=D_i$
$T1=(1, 5, 1, 5)$

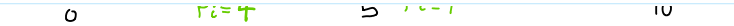
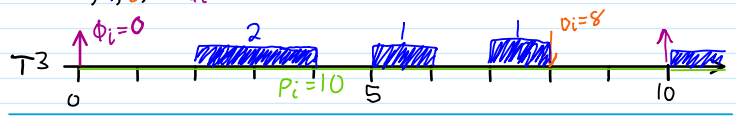
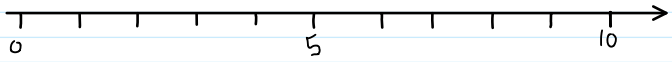


$T2=(4, 1) \rightarrow \phi_i=0$ & $D_i=P_i=4$



$T3=(10, 4, 8) \rightarrow \phi_i=0$ $\quad e_i=4=2+1+1$

$D_i=8$



$T3 = (10, 4, 8) \rightarrow \phi_i = 0$ $\quad e_i = 4 = 2+1+1$